

Coding & Solution

Date	30-06-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Yadlapalli-Sravya/EmotionDetector
Programming Language(s)	Python 3.11
Framework(s) Used	Streamlit, TensorFlow, Keras, Hugging Face Transformers
Key Features Implemented	User Registration & Login • Text Preprocessing using NLTK • Emotion Detection using BiLSTM & BERT

	• Mixed Emotion Detection with Confidence Score • Gemini AI Personalized Learning Support • Interactive Streamlit Dashboard • Plotly Analytics Dashboard • CSV Logging of User Interactions
Pending / Incomplete Features	None. All planned project features have been successfully implemented and tested.
Setup / Run Instructions	1. Install Python 3.11. 2. Install dependencies using pip install -r requirements.txt. 3. Configure the GOOGLE_API_KEY. 4. Place the trained model files in the models directory. 5. Run the application using streamlit run app.py.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes (after required dependencies and models are configured)
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **Emotion Detection & Learning Support Engine** is developed using a modular architecture that improves readability, maintainability, and scalability. The application integrates **BiLSTM** and **BERT** deep learning models for emotion classification and uses **Google Gemini AI** to generate personalized learning support based on the detected emotional state. The project follows clean coding practices with structured modules, meaningful naming conventions, proper error handling, and reusable components. The **Streamlit** interface provides an interactive user experience, while **CSV logging** and **Plotly analytics** enable performance tracking and visualization. The project can be further enhanced through multilingual support, Learning Management System (LMS) integration, and cloud deployment.